

No Data? No Problem: Robust Vision-Tabular Learning with Missing Values

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Abstract

Large-scale medical biobanks provide imaging data complemented by extensive tabular information, such as clinical measurements or demographics. However, this abundance of tabular attributes does not reflect real-world datasets, where only a subset of attributes may be available. This discrepancy calls for methods that remain robust to missing values at inference. To address this challenge, we propose RoVTL (Robust Vision-Tabular Learning), a framework designed to handle any level of tabular data availability, from 0% to 100%. RoVTL comprises two key stages: contrastive pretraining, where we introduce tabular attribute missingness as data augmentation to promote robustness, and downstream task tuning, where tabular missingness is complemented by a novel Tabular More vs. Fewer loss that ranks performance based on the amount of available tabular data. Combined with gated-cross attention fusion module, our tuning approach enables consistent performance across all tabular data completeness scenarios. We evaluate RoVTL on cardiac MRI scans from the UK Biobank, demonstrating superior robustness to missing tabular data compared to prior methods. Furthermore, RoVTL successfully generalizes to an external cardiac MRI dataset for multimodal disease classification, and extends to the natural images domain, achieving robust performance on a car advertisements dataset. The model weights and code are available at <https://github.com/marteczkah/RoVTL>.

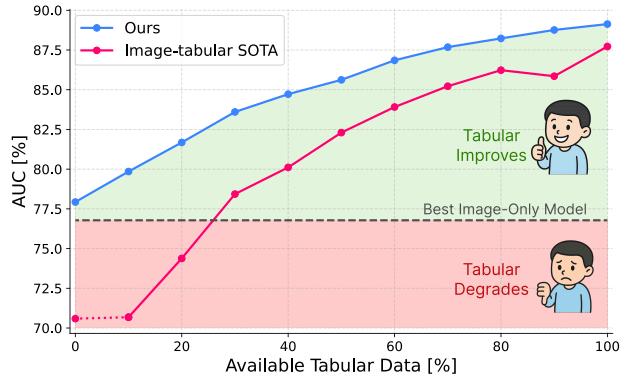


Figure 1. RoVTL consistently outperforms image-only inference (Hasny et al., 2026) and prior multimodal baselines (Du et al., 2024), addressing their key limitation of degraded performance in realistic clinical scenarios where less than 30% of tabular entries are available. The red dotted line indicates an edge case evaluated in our work but absent from the original paper (Du et al., 2024).

1. Introduction

Tables provide a structured way to represent diverse types of information, ranging from categorical to continuous variables, and thus capture aspects that are often not visible in images. When combined, tabular and imaging data offer complementary views: images encode spatial and morphological patterns, while tables provide contextual information. A valuable source of such multimodal data are large population studies, such as the UK Biobank (Sudlow et al., 2015) and the German National Cohort (GNC Consortium, 2014), which pair extensive tabular information with imaging data suitable for training vision-tabular models. However, even such studies frequently contain missing tabular entries, as complete data cannot always be collected from every participant. Furthermore, in real-world settings, it is often impractical to collect the same amount of tabular attributes (Dugdale et al., 1999), making it difficult to deploy models trained on those extensive datasets. This gap highlights the need for methods that can leverage the rich tabular information available during training while remaining robust when only limited to no tabular data are available.

Previous works have addressed those issues using two main approaches. The first, exemplified by MMCL (Hager et al., 2023) and TGV (Hasny et al., 2026), leverages both images

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and tabular data during pretraining, but relies solely on images for downstream tuning and prediction. This strategy addresses uncertainty of tabular data availability at inference. However, because not all tabular attributes have visual counterparts, valuable complementary information may be overlooked. The second strategy, adopted by methods such as TIP (Du et al., 2024), incorporates image and tabular data during both pretraining and inference, even when tabular entries are missing. However, this approach has not been evaluated in the extreme case where tabular data is absent. As a result, existing vision-tabular methods are typically designed for either image-only or joint image-tabular inference, without supporting both scenarios. However, in real-world settings, subjects may have complete, partial, or entirely missing tabular data. As shown in Fig. 1, this limited design of existing methods can cause the inclusion of tabular data to degrade performance relative to image-only models.

To fill the gap between the two strategies, we introduce RoVTL, **Robust Vision-Tabular Learning**, designed to deliver consistent performance across varying levels of tabular data availability. RoVTL maintains strong image-only capability while effectively leveraging tabular data when present. Our method is composed of two main steps: (1) We employ multimodal contrastive pretraining for our vision and tabular encoders, introducing *data missingness* as an augmentation strategy to improve robustness to incomplete tabular inputs. (2) We further employ the *data missingness* into the fine-tuning strategy, amplifying its effectiveness by proposing Tabular More vs. Fewer Loss (TabMoFe), which ranks the performance according to the number of present tabular attributes. The multimodal features are fused through a gated cross-attention mechanism that adaptively controls the influence of tabular data on the joint embeddings.

We conduct extensive experiments to evaluate the effectiveness of our proposed method on both classification and regression tasks across varying tabular data completeness scenarios. Specifically, we use cardiac MR images and clinical data from the UK Biobank (Sudlow et al., 2015) to assess performance on the challenging tasks of cardiovascular artery disease (CAD) classification and body mass index (BMI) regression. To assess the transferability of our pretrained encoders to settings with limited tabular attribute availability, we further evaluate cardiac disease classification on a public external dataset (Campello et al., 2021). In addition, we demonstrate that our framework seamlessly transfers to non-medical domains by evaluating it on a car advertisements dataset (Huang et al., 2022), showing its adaptability to natural vision-tabular settings.

In summary, our contributions are as follows. (1) We are the first to propose a method addressing the full spectrum of tabular data missingness, ranging from no to complete

tabular data. (2) We introduce RoVTL, featuring a novel data missingness augmentation approach and an effective downstream task fine-tuning strategy, enhanced by the proposed TabMoFe loss. (3) Experiments on medical and natural images datasets show that RoVTL surpasses the SOTA methods across varying tabular data availability scenarios, scoring the greatest improvement when little tabular data is available. (4) We show transferability to external cardiac dataset with reduced tabular attribute availability, highlighting its potential as a robust vision-tabular foundation model, and show the architectural generalization of RoVTL to the natural image domain.

2. Related Works

2.1. Learning with Tabular Data

Tabular data is a common modality in many disciplines, including finance and medicine. Traditional machine learning methods, such as decision tree models like XGBoost (Chen & Guestrin, 2016), achieve great performance on tabular data tasks that deep learning have struggled to surpass (Grinsztajn et al., 2022). While successful in unimodal setting, these models do not provide the capability to be used multimodally. In recent years, deep learning models, particularly those based on the transformer architecture (Vaswani et al., 2017), have started to rival and even outperform tree-based methods on sufficiently large datasets (Ye et al., 2024). Furthermore, the success of vision (Awais et al., 2025) and language foundation models (Zhao et al., 2023) has spurred interest in developing similar foundation models for tabular data. Models such as TabICL (Qu et al., 2025) and TabPFN (Hollmann et al., 2023) represent important advancements toward tabular foundation models. However, those models are often designed for a single task, which limits their scope and flexibility compared to foundation models in other domains, such as language and vision. TARTE (Kim et al., 2025) addresses these limitations by leveraging large knowledge bases for pretraining and encoding feature semantics through string representations. This design provides a general and transferable architecture, opening possibilities for effective multimodal learning with tabular data. We exploit this capability in our method.

2.2. Image-Tabular Data Learning.

End-to-End Approaches. Tabular data can improve visual learning by bringing complementary information that is not visible in the images, motivating the development of multimodal image-tabular methods. The approaches typically rely on fusing embeddings extracted from an image encoder and a separate tabular encoder (Duenias et al., 2025; Wolf et al., 2022; Spasov et al., 2019; Vale-Silva & Rohr, 2021; Duanmu et al., 2020; Borsos et al., 2024; Zheng et al., 2022). A different type of approach was presented in

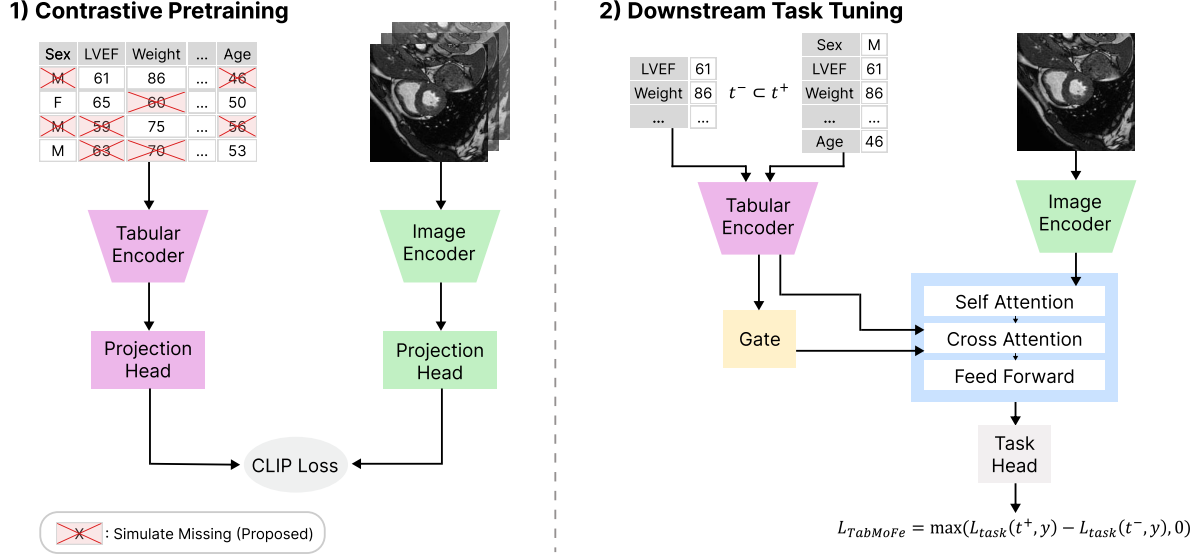


Figure 2. Overview of RoVTL. Our proposed framework, is composed of two main components: contrastive pretraining and downstream task tuning. In the pretraining stage, we introduce a novel tabular data missingness augmentation strategy, where random tabular attributes are dropped (red cross) to teach the encoder to handle missing entries. During downstream task tuning, two subsets of the tabular attributes t are used as input into the encoder and ranked by the proposed TabMoFe loss. Further, our multimodal fusion module weights the importance of the tabular data for the multimodal fusion. Not illustrated in the figure is DGL (Wei et al., 2025), which is integrated into our fine-tuning pipeline.

CHARMS (Jiang et al., 2024), where tabular data is only used at the training stage for selective knowledge transfer to allow image-only inference; and in STiL (Du et al., 2025), where semi-supervised learning is applied for image-tabular classification.

Pretraining-based Approaches. Multimodal self-supervised learning has achieved success in domains such as vision and language, as exemplified by CLIP (Radford et al., 2021). Similar pretraining strategies have been applied to image-tabular data learning. MMCL (Hager et al., 2023) was the first to use contrastive learning for vision-tabular data, using both modalities for the pretraining to achieve a powerful unimodal, image-only, inference. A similar approach was presented in TGV (Hasny et al., 2026), where tabular data similarity is used to guide the pair sampling process in contrastive learning. TIP (Du et al., 2024) was the first to use self-supervised pretraining for both image and tabular data inference, taking into consideration the missing values issue. However, TIP was not evaluated on the image-only edge case. Consequently, existing methods remain specialized either for image-only inference (MMCL, TGV) or joint image-tabular inference (TIP), with no model performing well in both scenarios. To bridge this divide, we introduce RoVTL, designed to handle the full spectrum of tabular data availability while maintaining strong image-only performance.

3. Methodology

3.1. Problem Definition

We consider a dataset consisting of image-tabular data pairs $\mathcal{D} = \{\{i, t\}_j, y_j\}_{j=1}^N$, where $\{i, t\}_j \in \mathcal{X}$ stand for the j -th image and tabular sample, respectively. Each tabular sample t_j may include a variable number of available attributes. The corresponding label $y_j \in \mathcal{Y}$ can represent either a categorical class (for classification tasks) or a continuous value (for regression tasks). We aim to find a mapping $f : \mathcal{X} \rightarrow \mathcal{Y}$ from the input space to the label space that remains robust to varying numbers of tabular attributes at inference, including the case where tabular attributes are completely missing.

3.2. Method Overview

RoVTL is organized into two main stages, as shown in Fig. 2. First, we perform contrastive pretraining by contrasting images with varying subsets of tabular attributes, a key aspect of our approach that enables the model to learn robust representations under different degrees of missingness. Second, we fine-tune the pretrained encoders for downstream tasks. For this, we supplement data missingness augmentation with the TabMoFe loss to rank performance across different attribute subsets, and employ a gated cross-attention-based fusion module in which the gate adaptively weights tabular features. To ensure effective multimodal optimization, we use disentangled gradient learning (DGL) (Wei et al., 2025).

3.3. Pretraining with Missingness Simulation

The primary objective of RoVTL is to encourage robustness across the full-spectrum of tabular data availability. We encourage such robustness at the pretraining stage by proposing to leverage a common challenge of missing entries in tabular data as a pretraining augmentation strategy. Our augmentation allows the tabular encoder to learn a representation that is robust to presence or absence of attributes at inference, without introducing attribute corruptions, such as sampling from empirical marginal distribution, that reduce the factuality of the pairs. Unlike prior methods (Hager et al., 2023; Du et al., 2024), we preserve the integrity of the true table, ensuring that the cross-modal alignment is anchored to authentic information.

Our multimodal contrastive pretraining follows the general setting of CLIP (Radford et al., 2021). Specifically, it uses two modality specific encoders, an image encoder E_i and a tabular encoder E_t , that are followed by projection heads f_i and f_t . For each image-tabular pair, we first apply augmentations to obtain $\tilde{i}$ and $\tilde{t}$. While we apply standard visual transformations, such as cropping and rotating, on the images, we define our tabular augmentations as a missingness simulation. Namely, given a sample’s full set of tabular attributes t^{full} with N_{full} entries, we randomly select a subset $t^{sub} \subseteq t^{full}$, where the size of t^{sub} is $[1, N_{full}]$, to serve as the input at each step, effectively simulating random entry missingness. In comparison to masking, which focuses on reconstructing missing values, this approach serves as a regularization technique that encourages the tabular encoder to learn meaningful representations, even when only partial attributes are available. The augmented inputs are then encoded by the respective encoders to produce embeddings $v_i = E_i(\tilde{i})$, $v_i \in \mathbb{R}^{d_i}$ and $v_t = E_t(\tilde{t})$, $v_t \in \mathbb{R}^{d_t}$. These embeddings are then passed through the projection heads to obtain the projections $z_i = f_i(v_i)$, $z_i \in \mathbb{R}^p$ and $z_t = f_t(v_t)$, $z_t \in \mathbb{R}^p$ that are used for the CLIP loss.

3.4. Combining Imaging and Tabular Features

After pretraining, the learned encoders are fine-tuned for downstream tasks. To generate multimodal embeddings, we use a gated vision-tabular fusion module. Since tabular attributes can vary in size and importance across samples, the amount of information in their embeddings can differ substantially. To adapt to this variability, we include a gating module, which allows the model to dynamically adjust the contribution of the tabular inputs to the multimodal embeddings, providing the flexibility to suppress uninformative or noisy signals. Formally, we extract the image and tabular features v_i and v_t from the pretrained encoders E_i and E_t . The embeddings are then projected into a shared latent space of dimension d via linear layers to obtain $\tilde{v}_i$ and $\tilde{v}_t$. The generated image embeddings $\tilde{v}_i$ are processed by a

multi-head self-attention block to capture the intra-image dependencies,

$$\hat{v}_i = \text{SelfAttention}(\tilde{v}_i) \quad (1)$$

Next, cross-attention is applied from the image tokens $\hat{v}_i$ to the tabular tokens $\tilde{v}_t$, where the image tokens serve as queries and the tabular tokens as keys and values. The [CLS] token from the tabular embedding $\tilde{v}_t$ is passed through a gating module implemented as a multilayer perceptron (MLP) with a final sigmoid activation, producing a weight scalar $w \in [0, 1]$. The final multimodal embedding is then computed as,

$$v_m = \hat{v}_i + w \odot \text{CrossAttention}(\hat{v}_i, \tilde{v}_t) \quad (2)$$

This multimodal representation is then passed through a feed-forward layer before being used by the downstream task heads for predictions.

3.5. Tabular More vs. Fewer Loss

The gated-cross attention module provides the architectural capacity to suppress the effect of noisy tabular attributes. To actively encourage this behavior at training, we extend the use of value missingness as a training-enhancement technique from pretraining to downstream task tuning. In contrast to other modalities, such as images or text, which can be defined as either present or absent, tabular data can have a variable number of attributes, making the notion of “missingness” more nuanced. Thus, applying modality missingness aware loss functions to tabular data requires specific adaptations. Crucially, the inclusion of more attributes does not inherently guarantee a less challenging prediction task as high-dimensional tabular settings can introduce noise and distractor variables, resulting in downgraded performance as more attributes are included.

We introduce TabMoFe loss, an adaptation to the SimMLM’s (Li et al., 2025) MoFe loss. MoFe encourages the model to achieve better performance when more modalities are available. We follow this notion, but instead of comparing modalities, the loss is designed such that the ranking is based on the number of given tabular attributes. That is, at each iteration we sample two sets of tabular attributes, t^+ and t^- . As different attributes carry different weights for the output, some being more important than others, we ensure that $t^- \subset t^+ \subseteq t^{full}$. Further, the number of attributes in t^+ is in range $[1, N_{full}]$, where N_{full} represents the maximum amount of tabular attributes available, and t^- is in range $[0, N_t)$, where N_t is the number of attributes in t^+ . Given this setting, we can assume that the performance of the model with t^+ should be better or equal to that of t^- as it should not perform worse when given more information. Given this, we define the TabMoFe loss as,

$$\mathcal{L}_{TabMoFe} = \max(\mathcal{L}_{task}(i, t^+) - \mathcal{L}_{task}(i, t^-), 0) \quad (3)$$

Then, the full multimodal task loss is defined as,

$$\mathcal{L}_{multi} = \mathcal{L}_{task}(i, t^+) + \mathcal{L}_{task}(i, t^-) + \lambda L_{TabMoFe} \quad (4)$$

where λ is introduced as a weighting factor. This objective directly addresses the robustness issues, encouraging the gated cross-attention module to prioritize informative attributes, while effectively filtering out the noisy variables that disrupt the performance monotonicity.

4. Experimental Setting

4.1. Datasets

UK Biobank (Sudlow et al., 2015) is a large-scale population-based cohort study. In this work, we use a subset of the biobank data consisting of pairs of short-axis cine Cardiac Magnetic Resonance (CMR) scans and tabular information incorporating attributes describing the demographics, lifestyle, and patient phenotype, among others. We use 39,975 pairs for training, 2,794 for validation, and 6,968 for testing. Our preprocessed CMR scans consist of 11 slices over 10 frames (sampled uniformly from the initial 50). The scans are cropped around the center of the heart to a size of 128×128 . Our tabular data consists of 129 attributes total, which include information such as sex, smoking status, or age. A full list of the used attributes and for which part of training they were used is available in Supplementary Material (SM) Table 5.

For the classification prediction target we use coronary artery disease (CAD) diagnosis based on the International Classification of Diseases (ICD10) codes, following the definitions in (Hager et al., 2023). For a full list of the codes see SM Sec. A.2. As CAD is underrepresented in the dataset (only 8% of the volunteer subjects have CAD) we prepare a disease balanced subsample of size 6,426 for the downstream task fine-tuning. The subsample is designed such that 50% of the data is CAD positive and 50% is CAD negative. We only consider the diagnosis reported before the scan as positive label. The second evaluation task is regression of body mass index (BMI), a continuous variable whose prediction benefits from tabular data. For the fine-tuning of the task we assemble a subsample of 5,000 image-tabular data pairs. The validation and test set are unaltered.

M&Ms is a multi-centre, multi-vendor, and multi-disease dataset from a cardiac image segmentation challenge (Campello et al., 2021) used as an external dataset to evaluate whether our method can generalize to cardiac data outside the UK Biobank. It consists of 150 training, 32 validation, and 106 testing short-axis cine CMR images coming from 5 different centers and 4 different vendors. The dataset also includes tabular attributes describing the patient phenotype and demographics, a full list is available in SM Table

6. We perform the same sampling of the volumes as for the UK Biobank, using 11 image slices over 10 frames, cropped around the center of the heart to a size of 128×128 . We use the dataset for cardiac disease classification, representing 4 different cardiac diseases and healthy samples.

Data Visual Marketing (DVM) dataset includes 1,451,784 images of cars at varying degree angles and their corresponding tabular attributes. We use the same preprocessing technique as (Hager et al., 2023), resulting in 70,565 training image-tabular pairs, 17,642 validation pairs, and 88,207 test pairs. The tabular attributes include fields on the cars’ length, height, or price. We evaluate the performance of our models on this dataset using model classification into 286 classes. As some of the attributes used in the pretraining can be used to directly identify the car model, such as the size of the car, we use a selection of attributes for the task tuning that ensures no information leakage between the training and test set. We note that our downstream attribute selection is more conservative compared to (Du et al., 2024) and consequently the performance metrics reported here differ from those reported in TIP. A full list of attributes used for the pretraining and fine-tuning is available in SM Table 7.

4.2. Implementation Details

All imaging encoders are based on the ResNet-50 (He et al., 2016) architecture, with a 3D version (Hara et al., 2018) used for the UK Biobank and 2D for DVM. The embedding size is set to 2048. For the tabular modality, we employ TARTE (Kim et al., 2025) as the encoder, initialized with its publicly released pretrained weights, with an embedding size of 768. The contrastive pretraining of RoVTL is implemented using Matryoshka Representation Learning (Kusupati et al., 2022), following TARTE with the projection head sizes of $\{64, 128, 256, 512, 768\}$. The baseline self-supervised pretraining methods are trained using their proposed settings, with a single projection head of size 128. The loss weighting parameter λ (Eq. 4) is empirically set to 0.05 for regression and 1 for classification. An analysis of the parameter λ is presented in SM Sec. C.2. We train and fine-tune the models for 100 epochs on UK Biobank and 500 epochs on DVM. The projection heads are removed for the fine-tuning. For further implementation details for RoVTL and the comparison models, see SM Sec. B. To improve the optimization of our method, we incorporate DGL (Wei et al., 2025) into RoVTL’s fine-tuning strategy. We provide details on DGL in SM Sec. B.3.

5. Results

5.1. Performance Comparison

We compare RoVTL against nine baseline methods, encompassing end-to-end, fusion, and pretraining-based ap-

Table 1. Performance comparison of RoVTl against baseline methods on UK Biobank CAD multilabel classification (AUC $\uparrow$), BMI regression (MAE $\downarrow$), and DVM car model multiclass classification (accuracy $\uparrow$). The performance is reported as a mean over tabular data incompleteness variants (0% - 100%). $\ast$: frozen backbones, $\circ$: trainable backbones.

	UK Biobank				DVM	
	CAD $\uparrow$		BMI $\downarrow$		Car Model $\uparrow$	
	$\ast$	$\circ$	$\ast$	$\circ$	$\ast$	$\circ$
<i>1) End-to-End Methods</i>						
Interact Fuse (Duanmu et al., 2020)	-	61.46	-	2.88	-	12.12
DAFT (Wolf et al., 2022)	-	68.59	-	2.41	-	42.83
<i>2) Fusion Methods</i>						
Max Fuse (Vale-Silva & Rohr, 2021)	74.45	81.38	2.91	1.60	90.14	86.05
Max Fuse* (Vale-Silva & Rohr, 2021)	77.61	82.90	1.91	1.59	94.25	94.65
Concat Fuse (Spasov et al., 2019)	75.13	76.33	3.05	1.77	90.02	86.37
Concat Fuse* (Spasov et al., 2019)	76.10	82.90	2.01	1.79	93.57	94.32
<i>3) Vision-Tabular Pretraining-based Methods</i>						
<i>Image Only Inference</i>						
MMCL (Hager et al., 2023)	73.52	75.43	2.24	2.22	91.64	94.06
TGV (Hasny et al., 2026)	73.22	76.78	2.52	2.28	92.52	94.23
<i>Image-Tabular Inference</i>						
TIP (Du et al., 2024)	78.08	80.49	2.16	2.13	94.73	95.31
RoVTl (Ours)	84.68	84.91	1.51	1.69	94.10	96.11

proaches. The supervised methods include Interact Fuse (IF) (Duanmu et al., 2020), and DAFT (Wolf et al., 2022), while the pretraining-based methods comprise TGV (Hasny et al., 2026) and MMCL (Hager et al., 2023), which perform vision-tabular pretraining followed by image-only inference. In contrast, TIP (Du et al., 2024) leverages images and tabular embeddings during both pretraining and inference. We also include multimodal fusion-based approaches Concat Fuse (CF) (Spasov et al., 2019) and Max Fuse (MF) (Vale-Silva & Rohr, 2021). To ensure proper comparison to those methods, we use our pretrained encoders for their training. Additionally, CF* and MF* denote variants trained under our proposed tuning setting to evaluate the transferability of our training to different networks. Unlike our approach, existing state-of-the-art vision-tabular methods are not designed to support the image only case, which is used for evaluations. Details of how we obtained the results for the image-only performance are described in SM Sec. B.4.

We evaluate all methods on two datasets, UK Biobank (Sudlow et al., 2015), where we perform multi-label CAD classification and BMI regression, and DVM car advertisement dataset (Huang et al., 2022), used for multi-class car model classification. Each model is trained in two configurations: trainable, where backbone weights are updated, and frozen, where backbones remain fixed. We report the performance as the average across varying levels of available tabular data (0%, 10%, ..., 100%), unless otherwise specified.

Table 1 shows the results of RoVTl compared with all baselines. For CAD classification, RoVTl consistently outperforms all other methods under both frozen and trainable configurations, surpassing TIP (Du et al., 2024) by 4.6% in

the frozen setting and 4.42% in the training setting. This improvement clearly demonstrates the benefit of incorporating robustness to missing entries during both pretraining and fine-tuning, unlike TIP, which only considers missingness during pretraining. Furthermore, CF* and MF* outperform their standard counterparts, proving the effectiveness of our downstream task tuning pipeline and showing that it is transferable to different networks and tuning methods. Similar trends are visible for BMI regression, showing our method can generalize to other tasks. While in the case of BMI regression, MF* achieves the best performance under the trainable backbones setting, the performance of our method under the frozen setting is still the best overall, suggesting the task particularly benefits from our pretraining strategy.

Lastly, to show that our method can generalize to 2D natural images, we evaluate it on the DVM car dataset for car model prediction. RoVTl outperforms the other baselines, achieving the best performance among all the settings. This result shows that RoVTl can also be applied to non-medical domain 2D imaging data without any adaptations. Notably, UK Biobank predictions involve over 80 attributes, compared to only 10 in DVM. While the relative improvement of RoVTl is more notable in the UK Biobank, it still benefits DVM, demonstrating the model’s effectiveness even with a small tabular attribute set.

5.2. Robustness to Missing Entries

The main objective of our method is to ensure robust performance across all levels of tabular data missingness, from no attributes to full availability. We visualize the performance of our method and the best performing image-tabular

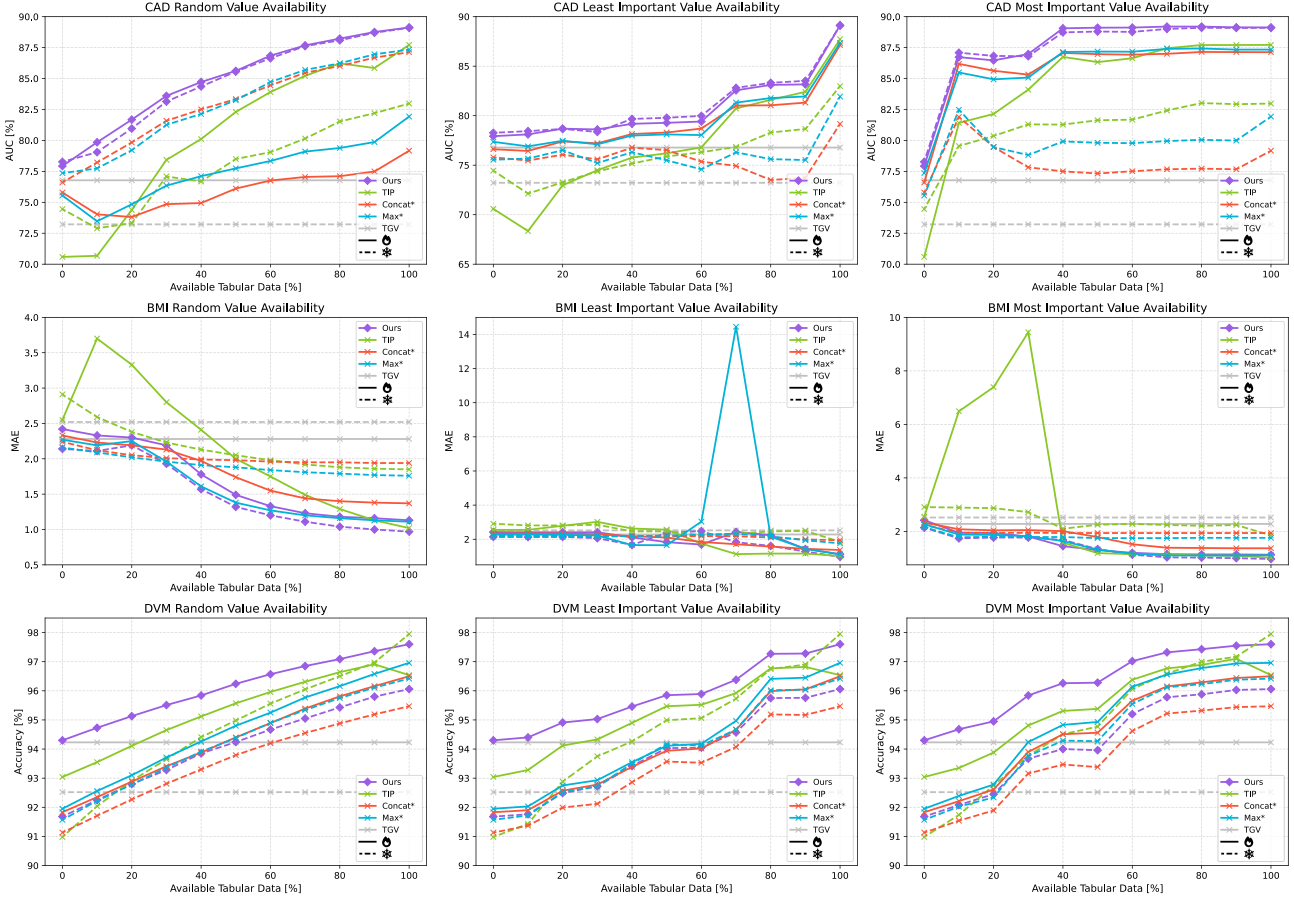


Figure 3. Model performance under different data availability scenarios from 0% to 100% tabular data used as input. CAD is evaluated using AUC ($\uparrow$), BMI using MAE ($\downarrow$), and DVM model classification using accuracy ($\uparrow$). $\ast$: frozen backbones, $\bullet$: trainable backbones.

baselines, TIP (Du et al., 2024), CF* (Spasov et al., 2019), and MF* (Vale-Silva & Rohr, 2021), and image-only TGV (Hasny et al., 2026) in Fig. 3. The performance of the other methods, including a tabular-only baseline, can be found in SM Sec. C.1. We show performance under three data missingness scenarios to account for different notions of missingness: random value, representing sample-level missingness, and least important (LI) and most important (MI) attributes, which account for column-level missingness. The LI and MI attributes are determined using random forest classifiers and regressors (Liaw et al., 2002), following (Du et al., 2024). While these attributes may not fully reflect each model’s notion of importance, this approach standardizes features across methods for fair comparison. RoVTL shows robustness to all presented tabular data missingness scenarios, achieving higher image-only performance than unimodal methods and improved performance monotonicity against prior vision-tabular methods, showing an improved capability of suppressing noisy signals. Furthermore, the biggest performance difference between RoVTL and TIP (Du et al., 2024) occurs when little to no tabular data is available, once again showing the importance of incorporating data

missingness strategies not only at pretraining, but also at downstream task tuning.

5.3. The Effect of TabMoFe Loss on Gating Behavior

Gated cross-attention is employed in RoVTL as a multi-modal fusion method, where the gate adjusts the impact of tabular data on the predictions. We evaluate how the TabMoFe loss shapes the behavior of the gating mechanism in Fig. 4. The MI and LI attributes are based on model’s respective attention values (SM Sec. C.3). Without our loss, the gate lacks a clear prioritization pattern, mostly focusing on the number of attributes, but not their importance. In contrast, the ranking loss enables the gate to understand the quality of the attributes. For MI attributes, the gate assigns its highest weights to the top 10% and scales back as less informative features are added to mitigate noise. For LI attributes, the gate remains low initially, only increasing as higher-quality information is included. Under random missingness, gate values stabilize and scale with data availability. These patterns show that the TabMoFe improves the gate’s ability to prioritize informative features and encourages per-

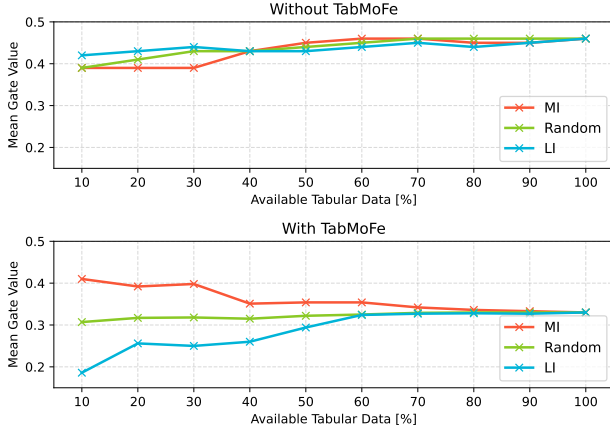


Figure 4. Mean cross-attention gate output as a function of tabular data availability under different missingness strategies, with and without the TabMoFe loss. LI: least important, MI: most important.

Table 2. Evaluation of different tabular data augmentation techniques for image-tabular pretraining. CAD is evaluated using AUC ($\uparrow$), and LVEF using MAE ($\downarrow$). $\ast$: frozen backbones, $\bullet$: trainable backbones.

	Image-Tabular		Image Only	
	CAD $\uparrow$		LVEF $\downarrow$	
	$\ast$	$\bullet$	$\ast$	$\bullet$
w/o Augmentations	83.49	83.44	3.13	3.01
Corrupted (MMCL (Hager et al., 2023))	81.07	83.59	3.26	2.99
Gaussian Pert.	71.60	78.64	4.45	3.14
Missing (Ours)	84.68	84.91	3.09	2.99

formance monotonicity.

5.4. Missingness as an Augmentation Technique

To prepare the tabular encoder for handling missing tabular entries, we propose simulating tabular data missingness as an augmentation technique for the pretraining. We compare the proposed missingness augmentation to value corruptions using Gaussian perturbations, sampling from feature’s marginal distribution, which has been applied in vision-tabular learning before (Hager et al., 2023), and no augmentations. The results are presented in Table 2. TARTE (Kim et al., 2025) is used as a tabular encoder for all the techniques. We evaluate the performance of the acquired backbones on CAD classification with both image and tabular data and for left ventricular ejection fraction (LVEF) regression using images only. LVEF, being image-derived, does not require tabular data for prediction and thus serves as an ideal target for our unimodal experiment. Our augmentation approach is the only one that consistently improves the no augmentation baseline across all the tasks. Notably, in a frozen setting all the value corruption-based methods fail to reach performance of the no augmentation setting. We attribute this behavior to the misalignment induced by

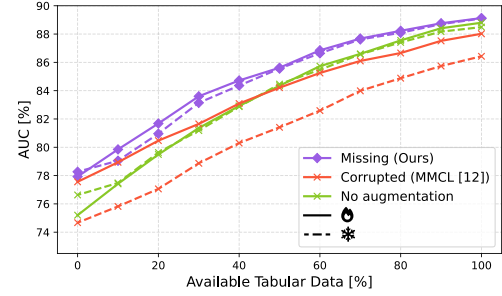


Figure 5. The effect of different tabular augmentation techniques used at pretraining on the result of CAD classification under random value missingness. $\ast$: frozen, $\bullet$: trainable backbones.

Table 3. The results on the multiclass cardiac disease classification on an external cardiac dataset called M&M’s (Campello et al., 2021). The results are reported using AUC. $\ast$: frozen, $\bullet$: trainable backbones.

	Disease Classification $\uparrow$	
	$\ast$	$\bullet$
Interact Fuse (Duanmu et al., 2020)	-	62.63
DAFT (Wolf et al., 2022)	-	57.45
MMCL (Hager et al., 2023)	64.93	65.37
TGV (Hasny et al., 2026)	51.59	63.01
RoVTl (w/o pretraining)	-	66.50
RoVTl (Ours)	76.43	77.56

corrupting the tabular attributes. While attribute corruption has been successful in unimodal tabular contrastive learning (Bahri et al., 2021), its application in vision-tabular learning can result in non-factual pairs, where the clinical information in the table no longer represents the ground truth of the patient. This is particularly harmful when applied to critical attributes like sex, smoking status, or disease labels, which fundamentally shift a patient’s profile and degrade the learned representations. In contrast, our missingness augmentation provides partial but truthful information, ensuring that cross-modal alignment remains anchored to authentic data, resulting in better performance. Furthermore, as illustrated in Fig. 5, our augmentation is especially beneficial at low tabular availability, confirming its role in promoting robustness to missing data.

5.5. Transferability to Attribute-Constrained Data

Development of tabular foundation models such as TARTE (Kim et al., 2025) can pave the way for multimodal vision-tabular models. We explore the potential of this idea by evaluating our vision and tabular encoders pretrained on the UK Biobank (Sudlow et al., 2015) dataset on an external cardiac dataset, M&M’s (Campello et al., 2021), for the task of cardiac disease classification. This dataset reflects the challenge of the discrepancy in the number of available tabular attributes in the clinics compared to Biobank, with M&M’s consisting of 16 attributes, compared to the 129

Table 4. Ablation study showing the effectiveness of each part of our method. The element listed as *w/o* is the one being removed. CAD is evaluated using AUC, while BMI using MAE. ✱: frozen backbones, ♡: trainable backbones.

	CAD $\uparrow$		BMI $\downarrow$	
	✱	♡	✱	♡
w/o pretraining	-	82.94	-	1.71
w/o gate	84.60	82.13	1.61	1.87
w/o downstream missingness	77.07	83.37	4.64	4.53
w/o TabMoFe	84.19	83.09	1.54	1.98
RoVTL (Ours)	84.68	84.91	1.51	1.69

used for the pretraining. Due to the rigid design of the tabular encoder in TIP (Du et al., 2024) we are unable to use TIP’s pretrained encoders for the comparison. Thus, we evaluate RoVTL against image-only inference methods such as MMCL (Hager et al., 2023) and TGV (Hasny et al., 2026), end-to-end supervised methods such as IF (Duanmu et al., 2020) and DAFT (Wolf et al., 2022), and a randomly initialized variant of our model. The results are presented in Table 3. RoVTL achieves the best performance, outperforming the randomly initialized model by 9.93 with frozen backbones and 11.06 with trainable backbone. These results highlight RoVTL’s potential to use the extensive knowledge from the UK Biobank and successfully transfer it into the clinical setting, where collecting extensive sets of tabular attributes is often impractical.

5.6. Ablation Study

Each element of our method was designed to ensure robustness to missing tabular data entries. To assess the contribution of each component, we conduct an ablation study in which one component is removed at a time. The results are summarized in Table 4. We use multilabel CAD classification and BMI regression for the ablations. RoVTL with all its components achieves the best results for all the tasks under both frozen and trainable backbones, clearly showing the effectiveness of our method. In addition, our proposed attribute missingness augmentation has the biggest impact in performance, demonstrating the importance of training under diverse tabular data availability scenarios.

6. Conclusion

We introduce RoVTL: Robust Vision-Tabular Learning, the first vision-tabular method, to our knowledge, that performs effectively across the full spectrum of tabular data completeness. RoVTL consists of two main components, contrastive pretraining and downstream task tuning. At pretraining, we propose tabular data missingness as an augmentation strategy, simulating incomplete entries to enhance the robustness of the tabular encoder. During downstream task tuning, data missingness is further enhanced with the proposed TabFoMe loss, which ranks performance based on available tabular at-

tributes, and a gated cross-attention module that adaptively adjusts tabular contributions. Our extensive experiments demonstrate that RoVTL outperforms prior methods, with the largest improvement under limited tabular data. Successful evaluation on external medical dataset highlights its potential for vision-tabular foundation models, while results on the 2D car dataset demonstrate its generalizability to multiple domains. Overall, RoVTL provides a generalizable and robust solution for vision-tabular learning, bridging the gap between image-only and multimodal inference and demonstrating effectiveness across the full spectrum of tabular data scenarios.

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Impact Statement

We introduce RoVTL, a method for robust vision-tabular learning across the entire spectrum of tabular data availability. The focus of our work is to allow biobank-scale learning, involving hundreds of tabular attributes, to be applicable in clinical practice, where such scale of tabular data is challenging to acquire. Our method can have a broad impact on decision making in clinical practice, allowing image-only models to be aided by tabular data, if such is available, without risking degraded performance. However, it is important to note that our pretrained encoders are trained using the UK Biobank cohort, which represents predominately white population. As a result, downstream use of these models may reflect or amplify demographic biases present in the training data, including potential racial bias. Careful evaluation across diverse populations is therefore necessary before deployment, and additional training or calibration using more representative cohorts may be required to ensure equitable performance.

Further, while our method is evaluated using different notions of missingness, namely random, least important, and most important, it is unclear how the performance would be affected for tasks where attributes might be Missing Not At Random (MNAR). In clinical practice, depending on a task or given hospital, there could be an underlying reason for the attribute to be missing. While we don’t find any MNAR attributes for our tasks, such evaluation is necessary before deploying the methods for tasks involving MNAR attributes.

Overall, RoVTL provides a path towards more robust vision-tabular learning, providing a foundation for developing systems that can adapt to real-world clinical constraints. We plan to extend this method to tasks that involve MNAR scenarios and further validate our models on varying demographics.

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A. Detailed Data Description

A.1. Tabular Attributes

Selection of the tabular attributes in our method is dependent on the training stage. We present comprehensive overview of the attributes, at which stage they are used, and their missingness rates in Table 5 for UK Biobank (Sudlow et al., 2015), Table 6 for M&Ms (Campello et al., 2021), and Table 7 for DVM (Huang et al., 2022). As the M&Ms dataset does not include missing attributes, we do not report the missingness rates. For the UK Biobank we also include the field IDs that are used to extract the attributes. The attributes that are marked as *extracted* are derived using automated cardiac MRI analysis (Bai et al., 2018).

A.2. ICD Codes

The labels for multilabel cardiovascular artery disease (CAD) classification task are defined using the ICD-10 codes of categories I20-I25. Diseases defined by the following codes are used as positive labels:

- Angina pectoris: I20.0, I20.2, I20.8, I20.9;
- Acute myocardial infarction: I21.0, I21.1, I21.2, I21.3, I21.4, I21.9;
- Other acute ischemic heart disease: I24.0, I24.8, I24.9;
- Chronic ischemic heart disease: I25.0, I25.1, I25.2, I25.3, I25.4, I25.5, I25.6, I25.8, I25.9.

We extract the date of each MR scan and consider the subject CAD positive only if the disease was reported prior to the scanning date. UK Biobank uses self-reporting of diseases and due to the nature of CAD, which can remain undiagnosed for years, it is possible that some of the healthy subjects do have CAD. Addressing the label uncertainty in UK Biobank is left for future work.

A.3. M&M’s Cardiac Disease Classification

The cardiac diseases included in the M&M’s dataset (Campello et al., 2021) differ from those used for the CAD classification in UK Biobank (Sudlow et al., 2015). M&M’s includes patients representing four cardiac diseases and healthy samples. The diseases are:

- Hypertrophic cardiomyopathy (HCM),
- Dilated cardiomyopathy (DCM),
- Hypertensive Heart Disease (HHD),
- Abnormal Right Ventricle (ARV).

B. Implementation Details

B.1. Image Augmentations

Image augmentations are applied using the torchvision library. Due to different data formats (2D vs. 3D) between DVM (Huang et al., 2022) and cardiac datasets, UK Biobank (Sudlow et al., 2015) and M&M’s (Campello et al., 2021), different augmentations are applied to them. For UK biobank and M&M’s the following augmentations are applied:

- RandomHorizontalFlip(probability=0.5),
- RandomResizedCrop(size=128, scale=(0.6, 1.0)),
- RandomRotation(degrees=45).

For DVM the following image augmentations are used:

- ColorJitter(brightness=0.8, contrast=0.8, saturation=0.8, probability=0.8),
- RandomGrayscale(probability=0.2),
- GaussianBlur(kernel_size=29, sigma=(0.1, 2.0), probability=0.5),
- RandomResizedCrop(size=128, scale=(0.6, 1.0), ratio=(0.75, 1.33)),
- RandomHorizontalFlip(probability=0.5),

B.2. Tabular Data Augmentations

Tabular Data Corruption. In our pretraining experiments assessing the impact of different augmentation strategies, we adopt two corruption schemes with an augmentation rate of 0.3, following prior works (Hager et al., 2023). The first corruption follows the marginal distribution sampling described in (Hager et al., 2023), where randomly selected entries are replaced with values observed elsewhere in the dataset. The second is a hybrid Gaussian corruption strategy: for a subset of features, numerical values are perturbed with additive Gaussian noise $\mathcal{N}(0, 0.5^2)$, while categorical features are subjected to random replacement with a different category observed in the column. In both schemes, the original value is replaced or modified to simulate data noise, and any pre-existing missing values are preserved to maintain the original sparsity pattern.

B.3. Tuning with Disentangled Gradient Learning

A common challenge in multimodal learning is under-optimization, where joint training can lead to lower performance compared to unimodal models. One of the techniques addressing this issue is disentangled gradient learning

Table 5. Tabular attributes extracted from the UK Biobank (Sudlow et al., 2015) and their corresponding field IDs. The ✓ symbol in the CAD or BMI column indicates that the attribute is used in the respective downstream task, while the ✗ symbol indicates that it is excluded from that task. All listed attributes are used during pretraining. Miss. describes the percentage (%) of values originally missing in the given column.

Tabular Feature	Field ID	Miss.	CAD	BMI	Tabular Feature	Field ID	Miss.	CAD	BMI
Alcohol drinker status	20117	0.7	✓	✓	Alcohol intake frequency	1558	0.7	✓	✓
Pulse rate automated	102	19.05	✓	✓	Augmentation index for PWA	12681	10.2	✓	✓
Average heart rate	22426	20.8	✓	✓	Basal metabolic rate	23105	5.2	✓	✓
Medication for cholesterol, blood pressure or diabetes	6177	7.4	✓	✓	Body fat percentage	23099	5.2	✓	✓
Body mass index	21001	3.2	✓	✗	Body surface area	22427	20.9	✓	✓
Cardiac index	22425	20.9	✓	✓	Cardiac index during PWA	12702	17.4	✓	✓
Cardiac operations performed	20004	26.4	✓	✓	Cardiac output	22424	20.8	✓	✓
Cardiac output during PWA	12682	13.6	✓	✓	Central augmentation pressure during PWA	12680	10.8	✓	✓
Central pulse pressure during PWA	12678	10.7	✓	✓	Central systolic blood pressure during PWA	12677	10.7	✓	✓
Cholesterol lowering medication regularly taken	6177	52.1	✓	✓	Current tobacco smoking	1239	7.4	✓	✓
Diastolic blood pressure automated	4079	19.0	✓	✓	Diastolic blood pressure manual reading	94	92.7	✓	✓
Diastolic brachial blood pressure during PWA	12675	10.7	✓	✓	Duration of heavy DIY	2634	53.7	✓	✓
Duration of moderate activity	894	10.2	✓	✓	Duration of strenuous sports	10010	87.8	✓	✓
Duration of vigorous activity	914	32.6	✓	✓	Duration of walks	874	2.3	✓	✓
Duration walking for pleasure	981	19.2	✓	✓	End systolic pressure during PWA	12683	10.9	✓	✓
End systolic pressure index during PWA	12684	13.6	✓	✓	Ever smoked	20160	1.1	✓	✓
Exposure to tobacco smoke at home	1269	2.7	✓	✓	Exposure to tobacco smoke outside home	1279	2.7	✓	✓
Falls in the last year	2296	0.7	✓	✓	Frequency of other exercises in last 4 weeks	3637	45.6	✓	✓
Heart rate during PWA	12673	10.2	✓	✓	Height	12144	0.0	✓	✗
Hip circumference	49	2.9	✓	✓	Hormone replacement therapy medication regularly taken	6153	48.6	✓	✓
Mean arterial pressure during PWA	12687	10.7	✓	✓	Number of beats in waveform average for PWA	12679	10.2	✓	✓
Days/week moderate activity	884	0.7	✓	✓	Days/week vigorous activity	904	0.7	✓	✓
Days/week walked	864	0.7	✓	✓	Overall health rating	2178	0.7	✓	✓
P duration	12338	16.5	✓	✓	Pace	3079	2.9	✓	✓
Pack years adult smoking as proportion of lifespan exposed	20162	76.5	✓	✓	Pack years of smoking	20161	76.5	✓	✓
Past tobacco smoking	1249	2.7	✓	✓	Peripheral pulse pressure during PWA	12676	10.7	✓	✓
Pulse wave Arterial Stiffness index	21021	16.0	✓	✓	QRS duration	12340	12.1	✓	✓
Shortness of breath walking on level ground	4717	0.7	✓	✓	Sleep duration	1160	0.7	✓	✓
Sleeplessness / insomnia	1200	0.7	✓	✓	Smoking/smokers in household	1259	2.7	✓	✓
Stroke volume during PWA	12686	13.5	✓	✓	Systolic blood pressure	4080	19.0	✓	✓
Systolic brachial blood pressure during PWA	12674	10.7	✓	✓	Total mass	23283	24.6	✓	✓
Total peripheral resistance during PWA	12685	10.9	✓	✓	Ventricular rate	12336	12.1	✓	✓
Waist circumference	48	2.9	✓	✓	Weight	21002	3.1	✓	✗
Whole body fat-free mass	23101	5.2	✓	✓	Whole body fat mass	23100	5.3	✓	✓
Whole body water mass	23102	5.2	✓	✓	Worrier / anxious feelings	1980	0.7	✓	✓
Date of birth (Age)	33	0.0	✓	✓	Sex	31	0.0	✓	✓
Heart Failure (I50)	13135	0.0	✗	✗	Atrial Fibrillation and Flutter (I48)	13135	0.0	✗	✗
Nonrheumatic Aortic Valve Disorders (I25)	131324	0.0	✗	✗	Other Diseases of Pericardium (I31)	131316	0.0	✗	✗
Paroxysmal Tachycardia (I47)	131348	0.0	✗	✗	Acute Pericarditis (I30)	131314	0.0	✗	✗
Atrioventricular and Left Bundle-Branch Block (I44)	131342	0.0	✗	✗	Nonrheumatic Mitral Valve Disorders (I34)	131322	0.0	✗	✗
Acute and Subacute Endocarditis (I33)	131320	0.0	✗	✗	Other Cardiac Arrhythmias (I49)	131352	0.0	✗	✗
Cardiomyopathy (I42)	131338	0.0	✗	✗	Acute Myocarditis (I40)	131334	0.0	✗	✗
Other Conduction Disorders (I45)	131344	0.0	✗	✗	Cardiac Arrest (I46)	131346	0.0	✗	✗
Nonrheumatic Tricuspid Valve Disorders (I36)	131326	0.0	✗	✗	Essential (Primary) Hypertension (I10)	131286	0.0	✗	✗
Hypertensive Heart Disease (I11)	131288	0.0	✗	✗	Hypertensive Renal Disease (I12)	131290	0.0	✗	✗
Hypertensive Heart and Renal Disease (I13)	131292	0.0	✗	✗	Secondary Hypertension (I15)	131294	0.0	✗	✗
Insulin-Dependent Diabetes Mellitus (E10)	130706	0.0	✗	✗	Non-Insulin-Dependent Diabetes Mellitus (E11)	130708	0.0	✗	✗
Other Specified Diabetes Mellitus (E13)	130712	0.0	✗	✗	Unspecified Diabetes Mellitus (E14)	130714	0.0	✗	✗
Angina Pectoris (I20)	131296	0.0	✗	✗	Other Acute Ischaemic Heart Diseases (I24)	131304	0.0	✗	✗
Chronic Ischaemic Heart Disease (I25)	131306	0.0	✗	✗	Myocardial Infarction (I21)	42000	0.0	✗	✗
Left Ventricular Ejection Fraction	Extracted	0.1	✓	✓	Left Ventricular End-diastolic Volume	Extracted	0.0	✓	✓
Left Ventricular End-systolic Volume	Extracted	0.0	✓	✓	Left Ventricular Stroke Volume	Extracted	0.0	✓	✓
Left Ventricular End-diastolic Mass	Extracted	0.0	✓	✓	Left Ventricular Cardiac Output	Extracted	0.0	✓	✓
Right Ventricular End-diastolic Volume	Extracted	0.0	✓	✓	Right Ventricular End-systolic Volume	Extracted	0.0	✓	✓
Right Ventricular Stroke Volume	Extracted	0.0	✓	✓	Right Ventricular Ejection Fraction	Extracted	0.1	✓	✓
Right Ventricular Cardiac Output	Extracted	0.0	✓	✓	Myocardial End-systolic Volume	Extracted	0.0	✓	✓
Myocardial End-diastolic Volume	Extracted	0.0	✓	✓					

DGL (Wei et al., 2025). We introduce this training strategy into our downstream task learning to ensure effective optimization of the unimodal models and the multimodal fusion module.

DGL explicitly decouples the optimization of modality encoders from that of the multimodal fusion module using two backward passes. In the first pass, a unimodal loss is computed for each modality, where separate task heads are attached to modality encoders,

$$\mathcal{L}_{unimodal} = \mathcal{L}_{task}(i) + \mathcal{L}_{task}(t) \quad (5)$$

This unimodal loss is used to update each modality encoder

independently. The gradients that accumulate on the multimodal fusion parameters during this step are removed, and the modality features are detached so that the upcoming multimodal loss does not propagate through them. In the second pass, the multimodal loss $\mathcal{L}_{multi}$ is computed and backpropagated only through the multimodal fusion module. This disentanglement resolves the gradient interference between unimodal and multimodal objectives, ensuring stable and balanced optimization across modalities.

B.4. Baselines

For the baselines, we follow the settings and specifications provided by the original authors. Any details specific to our implementations are described below.

Supervised baselines. The end-to-end supervised baselines, which include Interact Fuse (IF) (Duanmu et al., 2020) and DAFT (Wolf et al., 2022) are trained using the attributes as listed for each task in Tables 5- 7, without any pretraining including the remaining attributes. This approach ensures no leakage of task specific information, as pretraining incorporated attributes that are later used as labels for downstream tasks. The missing values are handled by concatenating the input attributes with a mask describing which attributes are present and which ones are missing. Zero imputation is used for the missing values.

TIP (Du et al., 2024). The original TIP pipeline does not include pretraining with labels, which was introduced in MMCL (Hager et al., 2023). To ensure proper comparison to other pretraining based approaches we use the same set of attributes, which also incorporate labels. For TIP, we use fixed masks at downstream task tuning that mask the attributes that should be excluded for the given downstream task prediction. As the TIP tabular encoder expects the attributes at certain order, we could not simply exclude the values and pad the remaining tokens, motivating the use of masking which was already implemented in the method as a way of marking missing entries. For the edge-case of tabular data being fully missing, we follow the typical TIP prediction pipeline, but mask all the tabular tokens.

Concat Fuse (CF) (Spasov et al., 2019) and Max Fuse (MF) (Vale-Silva & Rohr, 2021). We use two different multimodal embedding concatenation methods as baselines. Namely, CF where tabular and image embeddings are concatenated together, and MF, where element-wise max pooling is performed across two embeddings, selecting the larger values in each feature dimension to form the multimodal embedding. To ensure proper comparison, we use our pretrained image and tabular encoders for the two concatenation approaches. Furthermore, we compare RoVTL against those methods tuned using our downstream task tuning approach and the standard setting, without incorporating TabMoFe loss, missing value simulation, and DGL (Wei et al., 2025).

B.5. Downstream Task Tuning

We perform downstream task tuning under two settings, with frozen and trainable backbones. The multimodal fusion module is trainable in every setting and is added on top of the pretrained backbones. The Adam optimizer with a weight decay of $1e^{-4}$ and an effective batch size of 512 for DVM (Huang et al., 2022) and 256 for UK Biobank (Sudlow

Table 6. Full list of the tabular attributes from the M&M’s dataset (Campello et al., 2021). All attributes listed are used for the downstream task (cardiac disease classification), as denoted by ✓

Tabular Feature	Classification
Left Ventricular End-Diastolic Volume	✓
Left Ventricular End-Systolic Volume	✓
Left Ventricular Stroke Volume	✓
Left Ventricular Ejection Fraction	✓
Left Ventricular End-Diastolic Mass	✓
Left Ventricular Cardiac Output	✓
Right Ventricular End-Diastolic Volume	✓
Right Ventricular End-Systolic Volume	✓
Right Ventricular Stroke Volume	✓
Right Ventricular Ejection Fraction	✓
Right Ventricular Cardiac Output	✓
Myocardial End-Diastolic Volume	✓
Myocardial End-Systolic Volume	✓
Height	✓
Weight	✓
Sex	✓

Table 7. A list of tabular attributes from the DVM dataset (Huang et al., 2022). All the attributes are used for the pretraining. The attributes denoted by ✓ in the *Classification* column are used in the downstream task prediction (car model classification), while the ones marked by ✗ are excluded. Miss. describes the percentage (%) of values originally missing in the given column.

Tabular Feature	Miss. (%)	Classification
Manufacturer Name	0.0	✗
Car Model	0.0	✗
Color	8.1	✓
Registration Year	0.0	✓
Body Type	3.6	✓
Mileage Miles	0.4	✓
Engine Size	0.8	✓
Gearbox	0.1	✓
Fuel Type	0.2	✓
Price	0.4	✓
Engine Power	11.86	✗
Annual Tax	17.4	✗
Wheelbase	1.7	✗
Length	0.3	✗
Height	0.4	✗
Width	0.3	✗
Average Miles per Gallon	15.63	✗
Top Speed	16.25	✗
Seat Count	2.4	✓
Door Count	1.7	✓

et al., 2015) is used. We perform a learning rate sweep covering four values, $\{1e^{-3}, 3e^{-3}, 1e^{-4}, \text{ and } 3e^{-4}\}$, with the best one selected based on the validation set results. The tuning is performed for 500 epochs for the DVM dataset, 100 for M&Ms dataset (Campello et al., 2021), and 100 for UK Biobank. We use the Huber loss for regression (BMI regression), Binary Cross Entropy with Logits loss for multilabel classification (DVM car model prediction and M&Ms cardiac disease prediction), and Cross Entropy loss for multiclass classification (CAD classification on UK Biobank). The augmentation rate is set to 0.95.

C. Additional Experiments

C.1. Robustness to Missing Entries (Complete)

We provide full results for robustness to missing entries in Fig. 7. For clarity, we omitted baselines with the lowest performance in the main body of the paper. In addition to the baselines, we also include TARTE (Kim et al., 2025),

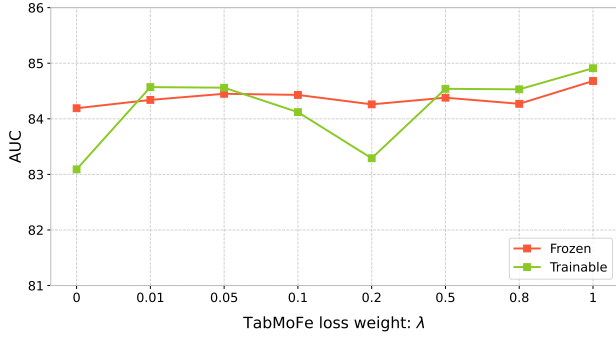


Figure 6. Impact of the TabMoFe loss weighting parameter λ on the CAD classification performance (AUC $\uparrow$).

the tabular encoder that we use for RoVTL, as a tabular-only model for comparison. As TARTE does not support native support for multilabel classification, we only include it as a baseline for BMI regression and car model classification. RoVTL outperforms the other methods on all tasks. Furthermore, incorporating our downstream fine-tuning setting to CF and MF improves the robustness of both of the methods to limited tabular data regimes, further showing the strength of our introduced training pipeline. RoVTL consistently outperforms TARTE, demonstrating greater robustness to missing data and underscoring the benefits of combining vision and tabular information.

C.2. λ -hyperparameter Analysis

We conduct an analysis of the effect of the TabMoFe loss weighting parameter λ on CAD classification performance. The results, illustrated in Fig. 6, show that aside from the $\lambda = 0$ baseline, where the TabMoFe loss is disabled, the model exhibits minimal performance variance across a wide range of values. These results demonstrate that our method is not sensitive to the specific selection of the λ hyperparameter, while confirming that the inclusion of the loss function consistently yields a positive effect on overall performance.

C.3. Interpretability

We evaluate the tabular attributes attention scores to better understand which features RoVTL uses to make predictions. The results are shown in Fig. 8 for CAD classification, Fig. 9 for BMI regression, and in Fig. 10 for car model prediction. To obtain the attention scores, the raw attention scores are extracted from the final cross-attention layer. The attention weights assigned to each attribute are then aggregated by first taking the mean over across all heads and then over the image tokens, which produces a single score for each tabular attribute. The aggregation is performed per class for classification and on full test set for regression. RoVTL takes advantage of the complementary information in the tabular attributes that is not directly visible on the image, such as

the engine size or mileage of the car, to make predictions. For BMI regression, attributes describing the body, such as the whole body fat mass or whole body fat-free mass, are attended to the most. This behavior confirms that the model does not treat the tabular data as auxiliary, but instead learns to rely on tabular features that provide predictive value beyond what is visible in the images, resulting in more robust and interpretable multimodal predictions.

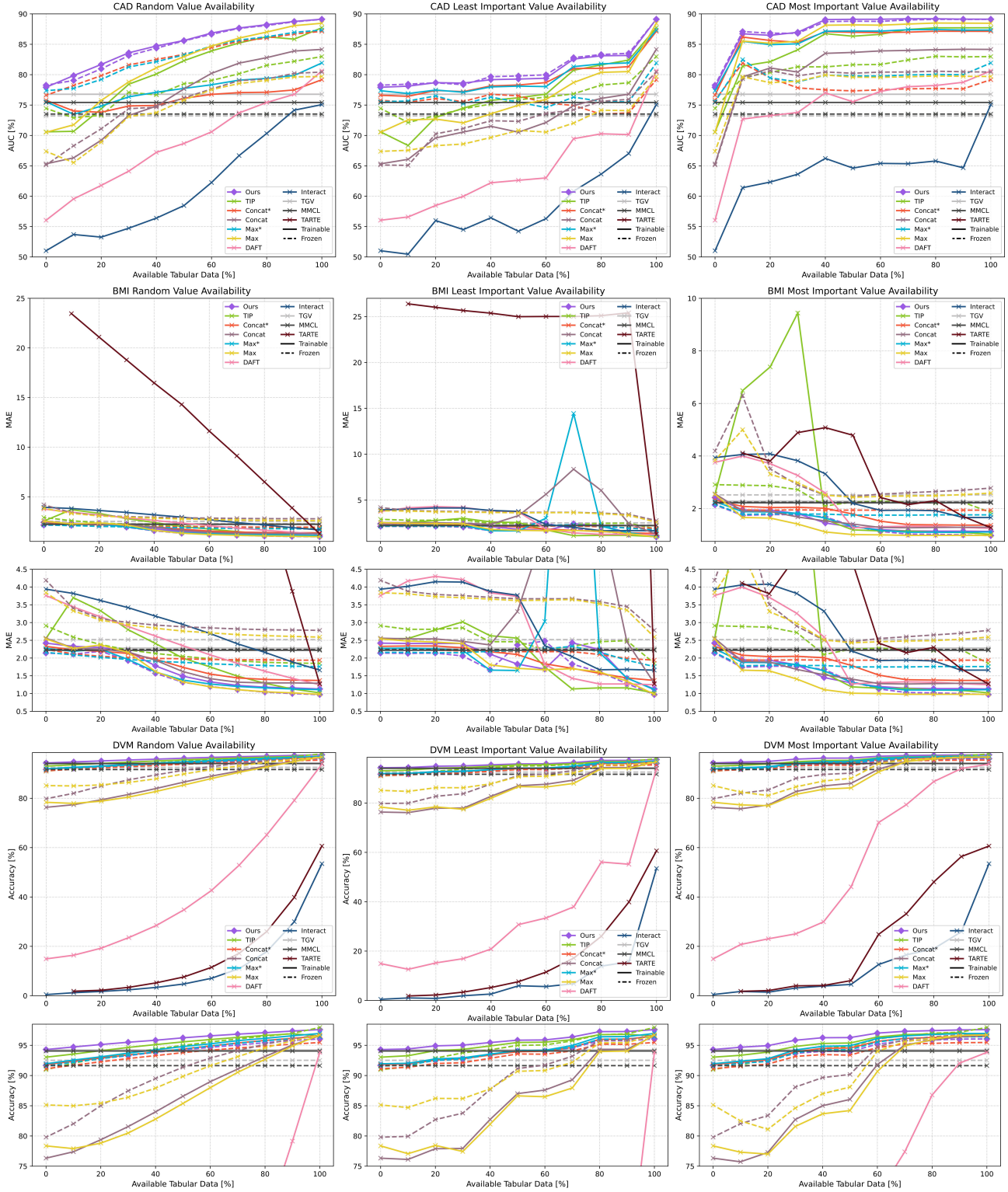


Figure 7. Model performance plotted over tabular data availability from 0% to 100%. CAD is reported using AUC ($\uparrow$), BMI using MAE ($\downarrow$), and DVM car classification using accuracy ($\uparrow$). For clarity, zoomed-in versions are provided for BMI regression and car model classification with DVM to highlight the performance differences between the models. * frozen backbones, $\bullet$ trainable backbones.

No Data? No Problem: Robust Vision-Tabular Learning with Missing Values

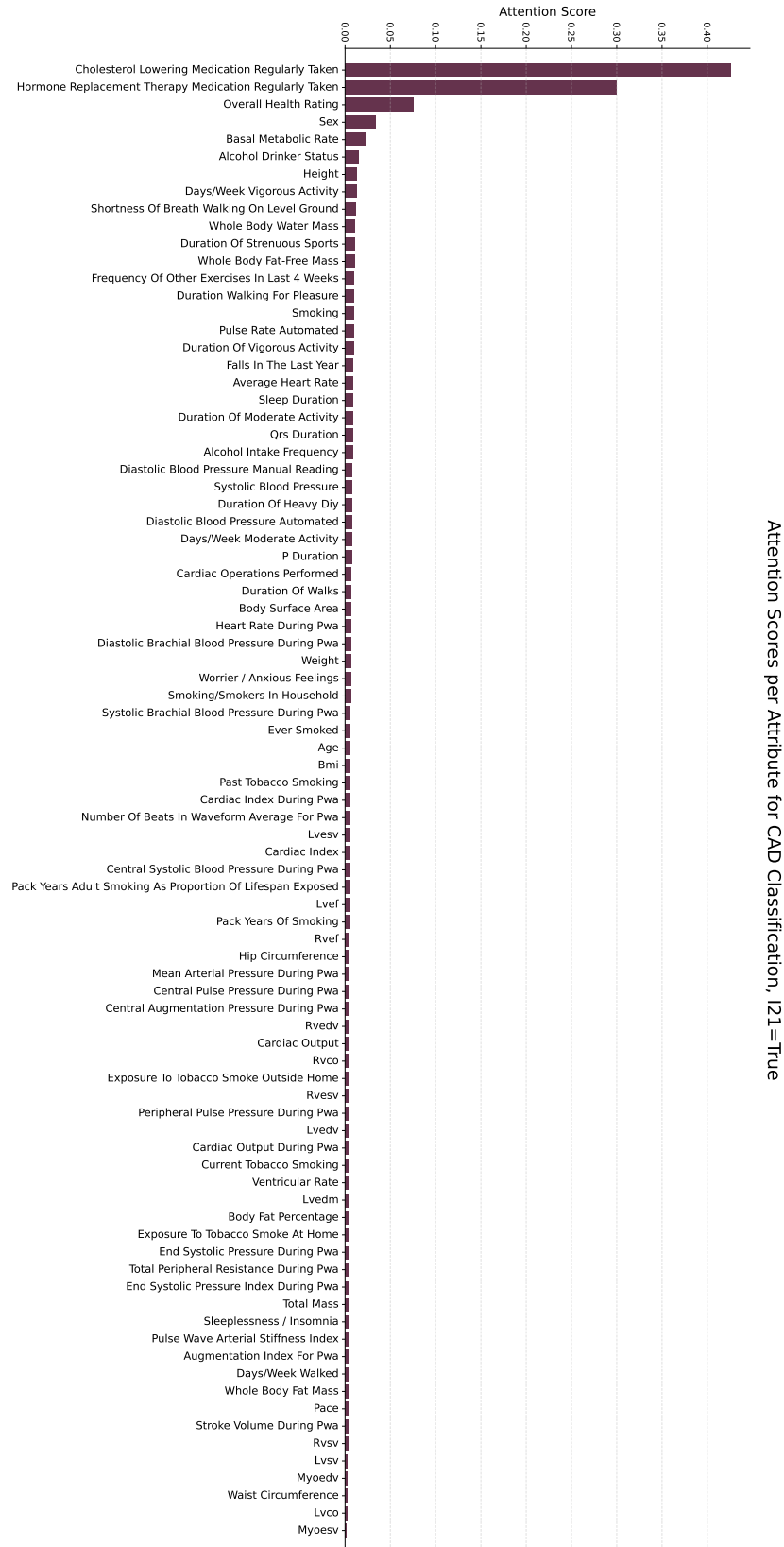


Figure 8. Attention scores per attribute for CAD classification with the UK Biobank.

No Data? No Problem: Robust Vision-Tabular Learning with Missing Values

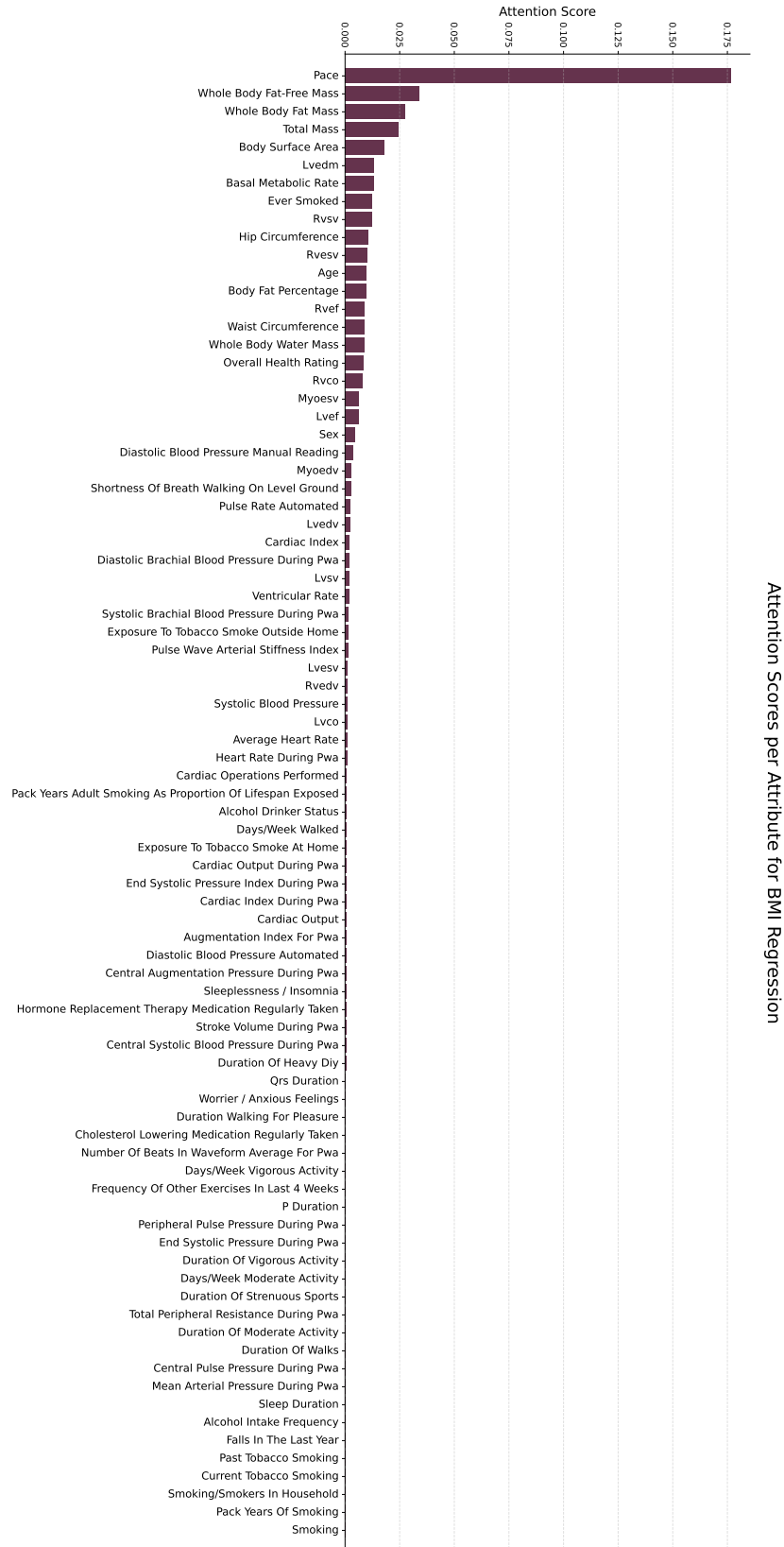


Figure 9. Attention scores per attribute for BMI regression with the UK Biobank.

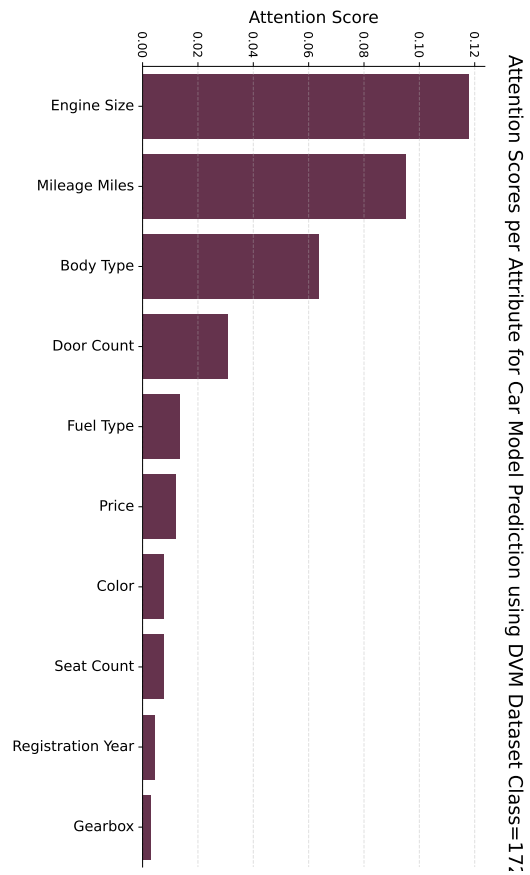


Figure 10. Attention scores per attribute for BMI regression with the UK Biobank.